

Project Motives and Objectives

Motives

The Redemption project starts off with the intention to replicate a quiz app that I was involved in making during a project in Year 3, but since I cannot figure out how to use websocket at that time, I ended up doing almost nothing in that project. Therefore, I want to remake this application to see what tech stack may need in order to complete this project.

Objectives

The objective of this project is not to make a product that gives me profit. This project is purely for educational purposes. It is overall a well-rounded application that allows me to utilize stacks like React, Express, Redis, [Socket.io](https://socket.io) etc. I didn't really expect to go this far in this project. But as time went by in making this project, more tech stack got involved. This document is used to document the technical details I did for this project. I can use this as a reference whenever I want to get back to this project later on.

MVP Description

The end product of Redemption only contains functionalities that are in the bare minimum. It contains feature of following categories:

- User account management
 - User login - basic login system with user entering their email and password
 - User account register - user can sign up a new account with their email address, an email will be sent to the registered email for account activation
 - Forgot password - if user forgot their password at some point, they can change it using their email, and enter the validation code they received in their email for identity validation
- Profile Page
 - History records - Those are records that the user has joined previously
 - Login calendar - It is a calendar that displays which date the user did log into their account
 - Score and statistics - Just some number metrics that shows the highest score and number of join history

- Problem Set CRUD
 - Users are able to create, read, update and delete problem sets under they account
 - User can edit the title and description of their problem sets
- Problems CRUD
 - Users can click into their problem sets and edit the problems within a problem set. User can edit the question type, question text, time allowed, option text (if MC), question order (drag & drop)
- Game Room Initialization
 - After user finish composing their problem set, they can initialize a game room which invites other users (both signed in and non signed in) to join the quiz room
 - Room Host can perform the following operations
 - Lock the room to prevent more people joining
 - Kick unwanted users within the room
 - Start a game once everything is ready
 - Terminate the room at anytime they want
- Joining an Existing Game Room
 - Support both signed in and non signed in users to join an existing room
 - Scores and history are not recorded for non signed in users
- In-Game Operations
 - In a game room, there are both hosts and normal participants. Operations are separated in 2 categories
 - Host
 - Terminate the on going game room
 - Inspect leaderboard
 - Kick existing player in the room
 - Participate in the game (host name will not be displayed in the leaderboard)
 - Participant
 - Participate in the game
 - Leave the room at anytime
- History Page
 - A page dedicated to store all the room the user joined in the past
 - User click can click into the records and inspect their past answers for a particular record (only available for completed status)
- History Content Review PAGE
 - This page is dedicated to allow users to navigate across their activities in the history record.
 - User can navigate forward and backward

Redemption Architecture

